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A Data-Driven MCDM Approach-Based Spherical Fuzzy Sets for Evaluating Global Augmented Reality Providers in Education

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ABSTRACT Amidst the surge in the popularity of augmented reality (AR), particularly in education due to the impact of the COVID-19 pandemic on remote learning, this research targets the challenge of efficient AR provider selection for educational use. The core objective revolves around introducing an encompassing decision support model, harnessing Spherical fuzzy sets (SFSs) to amplify decision-making. The study's initiation entails pinpointing eight pivotal criteria shaping AR provider selection via a stringent SF-Delphi procedure. The model duly factors each decision-maker's relative significance to ensure a fair and informed evaluation. The Spherical Weighted Arithmetic Mean (SWAM) operator amalgamates criteria weights, guaranteeing an all-encompassing and unbiased assessment. The ensuing step applies the Technique for Order of Preference by Similarity to the Ideal Solution (SF-TOPSIS) approach to systematically appraise and rank ten global AR providers. The systematic framework established in this study facilitates an impartial comparison among AR providers based on predetermined criteria. Through this research, a dual contribution is offered. Firstly, the proposed decision support model systematically and efficiently guides AR provider selection, effectively equipping decision-makers to navigate the intricacies of AR technology choices. Secondly, the research's impact extends to education, enriching the understanding of AR's applications. This knowledge fosters well-informed and innovative AR technology utilization, enhancing the overall learning experience. In conclusion, this study introduces a groundbreaking decision support model that leverages SFSs to guide the selection of educational AR providers. This structured framework empowers decision-makers and advances comprehension of AR dynamics, further driving strategic integration of AR technology in education.

INDEX TERMS Augmented reality; MCDM, Spherical fuzzy sets, Delphi, TOPSIS, Education.

I. INTRODUCTION

Augmented Reality (AR) technology has gained growing importance and recognition in various disciplines, most notably in educational settings [1]. This cutting-edge technology will evolve into an integral component of all human activity. Initially, AR transformed how technology is used and how we perceive things, altering the future of retail marketing systems and strategies by introducing previously imagined environments to customers [2]. Also, AR transformed the implementation and operation of businesses and processes, as well as had a considerable impact on our social lives, reshaping the entertainment industry by profoundly altering the gaming and filmmaking ecosystems [3].

AR technology is a powerful tool that can be used to enhance various experiences composed of two components: physical objects and digital content. It has been used to create immersive experiences in gaming, education, and marketing. In gaming, AR can be used to create a more realistic and engaging environment, allowing players to interact with virtual objects in the real world. In education, AR can create interactive learning experiences, allowing students to learn through visual and audio cues. In marketing, AR can create interactive campaigns, allowing consumers to interact with products and services more engagingly. AR technology can also create virtual tours, allowing users to explore places they may never have the chance to visit in person. By utilizing the power of AR, businesses can create more engaging and immersive experiences for their customers.

However, integrating them to create an interactive environment is a significant challenge, much more so when employed in educational contexts [4][5].

Additionally, adopting AR applications presents numerous problems, barriers, and difficulties due to the different ways AR technology delivers its solutions and its unique level of contextuality [6]. As a result, applying AR technology in any domain is challenging to accomplish with sufficient flexibility and adaptability. Consequently, these obstacles and significant issues have frequently been cited as reasons companies and individuals are skeptical about adopting AR technologies [6][7].

Academic institutions worldwide are more likely to employ massive-scale e-learning after the COVID-19 pandemic. Due to AR's ability to imitate the real world, it is envisaged that learning methodologies would incorporate some form of AR technology to address social distance limits and closures [8]. Thus, universities in developing countries are currently experiencing budgetary constraints that make it challenging to provide appropriate funding for educational development and quality [9]. Consequently, colleges and academic institutions can benefit from incorporating AR technology into the learning process since AR adds significant value and benefits to the learning landscape at a cheap cost. Moreover, numerous research studies have evaluated the most critical features of AR technologies across a range of application areas, most notably the learning and education landscape. This study aspires to make numerous contributions to contemporary literature. The adoption of AR inside educational systems continues to be a significant barrier. Considerable work still needs to introduce AR technology into mainstream educational settings [6]. Hence, this study contributes to contemporary literature by undertaking a more comprehensive investigation into the factors influencing the assessment and selection of AR software providers for education.

Among existing methods, multi-criteria decision-making (MCDM) is regarded as a practical approach based on a number of factors or criteria that influence the best solutions and are consistent with organizational strategies. Therefore, the selection of an appropriate AR application is an MCDM problem. Extensive study reviews are used to identify AR characteristics. The most critical characteristics are discussed in detail. Organizations needing AR solutions may use the proposed framework with the defined criteria list to select the best alternative. As ranking methods are increasingly improving, using an acceptable and effective MCDM method to determine alternatives' priority in AR selection is a requisite step.

On the other hand, fuzzy sets have received very little attention from the academic community, although some researchers have benefited to some extent from fuzzy sets when developing their MCDM models. In general, two-dimensional triangular fuzzy sets have addressed the inherent uncertainty in related problems. Apart from the numerous extensions of fuzzy linguistic term sets, several methods have been designed due to the diverse nature of MCDM problems. Some distinctive approaches among them

are VIšćKrite-rijumska Optimizacija I Kompromisno Rešenje (VIKOR), a technique for order of preferences by similarity to ideal solution (TOPSIS), complex proportional assessment (COPRAS), weighted aggregated sum product assessment (WASPAS), various versions of elimination and choice expressing the reality (ELECTRE) also recognized as outranking method, Multi-Objective Optimization based on ratio analysis plus full Multiplicative form (MULTIMOORA). For instance, Li et al. [10] applied VIKOR for machine tool selection using objective and subjective perspectives. Thus, the study of Mousavi-Nasab and Sotoudeh-Anvari [11] also used TOPSIS and COPRAS in the material selection problem. As indicated by Nguyen et al. [12], WASPAS was considered to rank government interventional strategies in the context of COVID-19. This method has been broadly applied to several practical problems [13]. The ELECTRE method is based on the study of outranking relations and uses concordance and discordance indexes to analyze the outranking relations among the alternatives [14]. Brauers and Zavadskas [15] proposed that MOORA be applied with the full multiplicative form; therefore, the MULTIMOORA method was created to rank alternatives.

Additionally, Liu et al. [16] introduced the ranking range-based approach, generating minimum and maximum ranking positions for alternatives via optimization models. The average ranking position is determined through Monte Carlo simulation, and these rankings are integrated to establish a novel composite ranking position for each alternative. Hwang and Yoon [17] proposed the Technique for Order of Preference by Similarity to the Ideal Solution (TOPSIS) method as a simple and valuable method for solving MCDM problems to select the best alternative with the shortest distance to the positive ideal solution (PIS) and the greatest distance to the negative ideal solution (NIS). Over the last few decades, various extensions of the TOPSIS method have been conducted under different fuzzy sets, such as ordinary fuzzy TOPSIS [18], interval-valued fuzzy TOPSIS [19], IF-TOPSIS [20], hesitant fuzzy TOPSIS [21], hesitant fuzzy linguistic TOPSIS [22], Neutrosophic TOPSIS [23], Pythagorean fuzzy TOPSIS [24], Interval-valued PF-TOPSIS [25].

Using spherical fuzzy sets in multi-attribute decision-making provides several advantages over conventional fuzzy set models. Spherical fuzzy sets allow for describing an expert's indeterminacy, membership, and non-membership degrees, all of which can be assigned within the unit sphere. This unique property distinguishes spherical fuzzy sets from other fuzzy set models and effectively deals with uncertainty and imprecision in decision-making problems. In real-world scenarios, evaluating alternatives in decision-making scenarios can be imprecise and unclear due to the instinctive nature of human cognition. Therefore, theories and methods have been developed to address decision-making under conditions of risk and uncertainty. The fuzzy MCDM technique has been utilized to measure and incorporate imprecision and ambiguity using fuzzy numbers in MCDM problems. Various extensions of general fuzzy sets have been developed, including type-two fuzzy sets, intuitionistic

fuzzy sets, neutrosophic sets, hesitant fuzzy sets, Pythagorean fuzzy sets, picture fuzzy sets, and spherical fuzzy sets. Recent extensions of conventional fuzzy sets with three-dimensional spherical geometry include SFSs [26], [27]. In SFSs, the sphere is considered a volume rather than a solid, allowing us to independently assign membership, non-membership, and hesitancy parameters in a larger domain than most other fuzzy sets. Interval-valued SFSs provide a greater capacity for modeling fuzziness with their interval-valued domain than a single point [28],[29].

Various decision-making techniques using spherical fuzzy sets have been proposed, including extending the classical TOPSIS model. In real-world problems, quantifying membership and non-membership as single values is often unreasonable. The spherical fuzzy TOPSIS method enables the incorporation of more comprehensive, hierarchically weighted criteria into the evaluation process for more accurate decisions. Systematically weighted criteria allow extensive evaluation of alternatives under the spherical fuzzy TOPSIS technique. This approach also permits using the same experts for criteria weighting and alternative assessment or different experts for each. In this study, the overarching goal is to develop a flexible framework for evaluating and selecting optimal AR solutions for education. The specific objectives are:

(i) Identify the most critical criteria for AR selection through literature review and expert validation.

(ii) Provide a research framework of the MCDM model to evaluate and rank AR provider alternatives under a spherical fuzzy environment.

As evidenced by the review of relevant literature, the selection of AR applications is one of the topics addressed in the academic literature. However, very few studies have examined this problem as a multi-criteria decision-making problem. Also, to the best of our knowledge, no similar research in the existing literature empirically examines AR indicators for evaluating and selecting AR providers in an education context using SF-TOPSIS. Therefore, this research addresses this gap and investigates vital questions, including:

(i) What are the most important criteria for evaluating and selecting AR providers for education?

(ii) How can the Spherical fuzzy MCDM method of Delphi and TOPSIS be validated for an AR provider selection problem in education?

This study makes several contributions to the literature to address the identified research gap and answer the key questions. Specifically, by developing and validating a spherical fuzzy MCDM methodology for AR provider evaluation and selection in education, the following contributions are made:

(i) Using spherical fuzzy sets to represent AR criteria evaluations to reduce subjectivity and uncertainty.

(ii) Aggregating expert opinions under a spherical fuzzy environment using the spherical weighted arithmetic mean operator to get crisp criteria values.

(iii) Establishing a technique to rank AR providers for education using SF-Delphi validated criteria. Criteria weights are calculated using the spherical weighted arithmetic mean operator to handle uncertainty. SF-TOPSIS ranks alternatives accurately.

The proposed spherical fuzzy MCDM approach enables modeling vagueness in assessments while reducing bias. Expert opinions are aggregated to determine salient criteria weights. The SF-Delphi and TOPSIS techniques are combined to evaluate and rank AR alternatives systematically. The study offers significant theoretical and practical contributions by developing and validating this methodology through a real-world example.

The remainder of this study will be as follows. Section two establishes the theoretical framework. Section three contains the research hypotheses and describes the technique used in the research. Section four summarizes the study's findings. Section five discusses the results of the research. The conclusion, theoretical contributions, practical implications, limitations, and future research are discussed in the final part.

II. LITERATURE REVIEW

A. AUGMENTED REALITY

Recently, AR has emerged as one of the most promising digital technologies, capturing the interest of many academics and practitioners. Above all, a wide variety of fields have adopted AR technology. Indeed, its use has grown in popularity across various educational contexts due to its benefits in boosting and improving learning processes and outcomes [30]. Additionally, AR technology is becoming more prevalent in the tourism and travel industries, and this technology has the potential to enhance travelers' experiences and expectations significantly. Also, research indicates that AR technology has been implemented in healthcare to increase efficiency and productivity and offer high-quality services [31], [32].

Additionally, AR technology has been adopted by the gaming and entertainment industries [33]. Recently, AR has been widely associated with the Internet of Things (IoT) as a feasible tool [34]. Finally, additional fields such as industrial production, advertising, engineering and construction, marketing and sales, and military technology have taken and utilized AR.

Nevertheless, numerous obstacles and crucial concerns have contributed to the widespread apprehension of most companies and individuals about embracing AR technologies, such as the cost of implementation, the potential for privacy issues, and the lack of understanding of the technology [6]. Thus, the lack of a clear roadmap for the future of AR and the risk of obsolescence have caused many to be hesitant to embrace the technology [35]. As indicated

by Ashtari et al. [36], the challenge of creating original, informative, rich, and exciting material has been a significant impediment to the technology's successful adoption by potential users, as people will eventually focus more on the content than on the technology itself. Finally, AR can contain terrifying and disturbing experiences and information that do not align with the target audience's demographics, tastes, and expectations. This may, to a degree, prevent people from establishing positive, purposeful behaviors toward technology acquisition and acceptance [37].

B. AUGMENTED REALITY IN EDUCATION

AR technology has become an increasingly popular tool in educational practices. AR can be used to create virtual classrooms, allowing students to interact with the environment more interactively and engagingly. For example, a virtual classroom could allow students to explore a 3D model of a historical site or a 3D model of a scientific experiment. This would enable students to understand the subject matter better and make the learning process more engaging [38]. Additionally, AR can create interactive learning experiences, such as virtual museum tours or interactive games that help students learn.

Also, AR technology also has the potential to create personalized learning experiences. For example, AR can create interactive lessons tailored to the individual student's learning style. This could help students learn more effectively and make learning more enjoyable [39].

Bazarov et al. [40] used an AR mobile application to engage students in an interactive lesson about electrical elements in a laboratory setting, and their findings indicated that the AR application provided more demonstrative and appealing content for students to learn independently, resulting in a more cost-effective and time-saving solution for teachers. Additionally, the AR application engaged pupils since they demonstrated "sympathy and affection" for augmented reality technologies.

Furthermore, Bulagang and Baharum [41] developed a mobile application using AR technologies to assist students in learning the foundations of computer networks. Students can manipulate virtual three-dimensional objects and interact with them by adding information such as labels.

According to Tekedere and Hoker [42], AR technology combines and integrates virtual things with real-world visuals via video, sound, photography, text, and 3D models, among other approaches. It is considered to be a subset of virtual reality. A critical distinction between the two is that, unlike VR, AR gives users a good interface for integrating the real and virtual worlds.

Cai et al. [43] indicated that enabling interaction between users and virtual items embedded in real-world settings results in natural and virtual interactions. In addition, the use of AR technology in sports broadcasting, theoretical immersion, physical fitness, and physical education is growing. Chin et al. [44] observed that AR technologies have high accessibility, interactive connectedness, and multi-dimensional data digitization, all necessary for bridging contemporary sports growth and technological improvement. Chang et al. [45] described how AR could be integrated into physical education classrooms by combining digital teaching materials (laws and notions of spatial skills) with virtual artifacts from 3D-assisted training resources (overview and execution of spatial skills). It enables learners to interpret digital three-dimensional interactive objects while studying, effectively retaining factual and procedural knowledge and boosting the interactive learning experience.

AR is a unique way of integrating digital content with the physical environment. Therefore, organizations need to leverage this new technology to train staff via self-service instructions in education, conduct real-time quality assurance sessions, and create targeted interactive advertising to increase brand loyalty.

A thorough analysis is essential before organizations can pick which AR software to purchase. Following selecting the most critical essential features, an MCDM procedure should be carried out. This study assists organizations and individuals in comprehending the critical criteria to consider when selecting an AR solution and provides a framework for selecting the optimal solution for their needs. In organizations' digital transformation journeys, the fuzzy framework is proposed to convert subjective human judgments into objective decision suggestions.

C. MCDM MODELS-BASED SPHERICAL FUZZY SETS

The MCDM procedure is a systematic method for identifying the optimal solution from a set of alternatives by comparing each alternative to a set of criteria [46]. Nonetheless, imprecision and uncertainty are two major concerns in MCDM problems. To address these issues, the Fuzzy MCDM technique, which measures and incorporates imprecision and ambiguity using fuzzy numbers, is utilized. Due to the instinctive nature of human cognition, evaluations of alternatives in real-world decision-making scenarios are frequently imprecise and unclear [47]. Therefore, the decision-making process under risks and uncertainty conditions has resulted in theories and methods [48]. The fuzzy set logic proposed by Zadeh [49] significantly impacts the solution of decision-making issues in uncertain

environments. Since 1965, many academics have expanded various extensions of general fuzzy sets. Zadeh [50] developed type-two fuzzy sets, Atanassov [51] developed intuitionistic fuzzy sets, Smarandache [52] developed neutrosophic sets, Torra [53] developed hesitant fuzzy sets, Yager [54] developed Pythagorean fuzzy sets, Cuong [55] developed picture fuzzy sets, and Mahmood et al. [26] and Kutlu Gundodu and Kahraman [27] proposed spherical fuzzy sets (SFSs). An expert's indeterminacy, membership, and non-membership degrees can be described using SFS, the latest extension of fuzzy sets. All three parameters may be assigned by experts as long as they remain within the unit sphere. This unique property distinguishes the spherical fuzzy set from conventional fuzzy set models [55]. The notion of SFSs is beneficial and effective for dealing with uncertainty and imprecision in multi-attribute decision-making issues [56]. In real-world challenges, quantifying membership and non-membership degrees in terms of a single value number is unreasonable. For this reason, Kutlu Gündodu and Kahraman [30] developed the concept of IVSFSs. The researchers have proposed several decision-making techniques in the literature by utilizing SFSs [57]–[59]. The classical TOPSIS model has been extended to include all fuzzy set types, including type-2 fuzzy TOPSIS, hesitant fuzzy TOPSIS, intuitionistic fuzzy TOPSIS, and Pythagorean fuzzy TOPSIS.

Intuitionistic fuzzy sets (IFS) and Pythagorean fuzzy sets (PFS) have limitations that render them inadequate for handling uncertainty in data and capturing all data. Specifically, there is no function to represent the hesitancy level of decision-makers [60]. Recently, SFSs have provided an effective method for determining ambiguity in information and better represent the opinion of decision-makers than existing fuzzy sets. SFS permits decision-makers to assign their hesitancy in the decision environment independently. Thus, SFS enables the decision-making process to be more intelligent, for example, comparable to human judgment, resulting in greater precision in determining weight and evaluating alternatives in complex decision-making problems [61]. SFS can be viewed as synthesizing neutrosophic, and PFS sets [6]. Decision-makers indicate their degree of hesitancy as membership degree and non-membership degree under a spherical fuzzy environment. Accordingly, SFS combines the benefits of other fuzzy sets and eliminates several drawbacks of neutrosophic and PFS sets. Kutlu Gündoğdu and Kahraman [62] introduced and employed the SF-TOPSIS approach to solving a multiple-criteria selection issue to determine the optimal location for an electric vehicle charging station. Mathew et al. [63] combined SF-AHP and SF-TOPSIS to select an advanced manufacturing system with six evaluation criteria and four options. Kahraman et al. [64] applied SF-TOPSIS to the problem of hospital location

selection. Nguyen proposed a method by integrating SF-Delphi with SF-TOPSIS methods for tour package selection in the context of Vietnam.

However, there are no studies using SF-Delphi and SF-TOPSIS to evaluate and select global providers of AR software for education. In the context of utilizing the Delphi method, a pivotal concern is attaining consensus. This issue warrants exploration, and the authors are recommended to draw insights from existing research. Notably, the literature review highlights two pivotal studies that shed light on the subject. To begin, Xiao [65] offers a consensus-focused approach to enhance the efficiency of Failure Mode and Effect Analysis (FMEA). This method integrates linguistic distribution among FMEA team members, enabling the derivation of ordinal classifications for potential failure modes (FMs). The study devises a multi-stage consensus optimization model with bounded confidence, facilitating consensus attainment. It introduces a maximum consensus optimization approach, generating adjustment suggestions to maximize consensus levels. In cases where predetermined consensus thresholds are not met, these suggestions guide the preference-modification of team members. A two-stage consensus optimization model is also introduced, catering to scenarios where consensus is achieved, facilitating preference modification among team members. In a parallel vein, Zhang et al. [66] contribute to the discourse on consensus in Group Decision-Making (GDM) by introducing Comparative Linguistic Expression Preference Relations (CLEPRs). This mechanism accommodates uncertain viewpoints and integrates multiple self-confidence levels, bolstering the integrity of decision-making processes.

Furthermore, the study extends CLEPRs to embrace personalized individual semantics, accompanied by an optimization model grounded in consistency-driven methodology. This model evaluates individual semantics within CLEPRs characterized by self-confidence levels, catering to diverse interpretations by decision-makers. An optimization-based consensus model is then presented to achieve a collective solution, strategically minimizing information loss between decision-makers preference relations and corresponding individual preference vectors.

Hence, this study aims to bridge this existing gap in the literature. In the realm of multiple-criteria decision-making, the selection of an appropriate methodology is crucial to ensure accurate and effective decision outcomes. With an array of approaches available, it is essential to justify the rationale behind the chosen methodologies meticulously. This motivation stems from recognizing that the efficacy of decision-making methodologies is contingent upon their alignment with the research's specific objectives and the unique challenges presented by the study's context. In this study, the motivation for selecting the Delphi method, TOPSIS, and SFSs lies in their distinct attributes that harmoniously address the multifaceted requirements of AR

provider selection for educational purposes. Each of these methodologies brings a unique set of strengths that, when integrated, are poised to provide a comprehensive and accurate solution to the research problem. Firstly, the Delphi method's utilization is driven by its ability to harness collective expert insights and achieve consensus in complex decision scenarios. This method's iterative process accommodates diverse viewpoints, rendering it suitable for synthesizing preferences and opinions from various stakeholders involved in AR provider selection. TOPSIS, on the other hand, is adept at handling multi-criteria decision-making by quantitatively evaluating alternatives' proximity to the ideal solution. Its application in this study is warranted due to the need for objective and systematic comparison among AR providers, considering various criteria simultaneously. Incorporating Spherical fuzzy sets complements the decision-making process by addressing uncertainties and subjectivity inherent in linguistic expressions and preferences. This is especially pertinent in AR provider selection, where imprecision and ambiguity can arise. This study aims to achieve a holistic and refined approach to AR provider selection by synergistically employing these methodologies. Thus, the motivation to choose the Delphi method, TOPSIS, and Spherical fuzzy sets arises from their compatibility with the study's objectives and their capacity to navigate the intricate landscape of educational AR provider selection, contributing to informed and adequate decision-making in this domain.

The previous parts discuss the impact of AR applications on educational digitalization strategy and methodology. All possible challenges should be considered before such essential software can be deployed on-site. Several challenges were observed: costly authoring, item tracking, virtual item placement issues, latency, lack of interactivity, and accuracy restrictions [64]. As a result, choosing the right AR provider will help organizations reduce the total number of product defects. Based on experts' opinions and the related references from the literature, the most relevant criteria for the evaluation process are summarized in **Table 1**.

TABLE 1. RELEVANT CRITERIA

AR criteria	References
Hardware Support	Zigart and Schlund (2020) [67]
Content Support	Alcácer and Cruz-Machado (2019) [68]
Authoring	Zigart and Schlund (2020) [67]
Reputation and References	Thanaraksakul et al. (2009) [69]
Tracking	Uva et al. (2018) [70]
Registration	Ong et al. (2008) [71]
Integration and Real-time data	Alcácer and Cruz-Machado (2019) [68]
Architecture & Deployment	Chi et al. (2013) [72]
Cost	Dieck et al. (2016) [73]

(HS) Hardware feature: Complexity AR technology can be divided into five categories, including Head-Mounted Displays (HMDs), Hand-Held Displays (HHDs), monitors, projectors, and 2D smart glasses, based on the four primary requirements for defining the characteristics of AR hardware [67], [74] as shown in **Table 2**.

TABLE 2. EXISTING AR HARDWARE FEATURES

Features	HMDs	HHDs	Monitors	Projectors	2D Smart Glasses
Mobility	γ	γ			γ
3D Space	γ	γ	γ	γ	
Registration					
Hands-free	γ		γ	γ	γ
Real-time data transmission	γ	γ	γ	γ	γ

(CS) Content feature: Virtual content in AR applications can be either two-dimensional (2D) or three-dimensional (3D) [68]. Depending on the manufacturer's preferred software, various 3D model formats are available. Support for native formats is essential for delivering a higher quality. AR apps should support a variety of multimedia formats and types to facilitate authoring [69].

(AU) Authoring feature: Authoring is a rapidly expanding subject within the AR domain, requiring time demanding and a high level of authoring knowledge [67]. Therefore, the optimal AR technology should support authoring by requiring no software coding, facilitating integration and navigation within the experience, and enabling user feedback [70].

(RR) Reputation and References: An organization's standing and reputation is based on how well its sales volume compares to its rivals in the same industry [69]. Research the AR provider's reputation in the market and look for customer references, case studies, or testimonials. This provides insights into their expertise, reliability, and success in delivering AR solutions for other organizations [74].

(TR) Tracking feature: The tracking is intended to be easily recognizable by the hardware. The objective is to be precise while enhancing the virtual content with the appropriate real-world location. The current trend in AR applications is marker-less tracking or free tracking [70] by converting 3D models to 2D sketches and using edges, corners, and other dimensional parameters to launch the experience when the model is discovered in the environment [75]. AR applications' success and accuracy depend on their tracking method. Makers should experiment with different monitoring methods. Consequently, the AR app should

support many tracking methods for various causes and scenarios.

(RE) Registration feature: Registration creates a virtual coordinate space and positions virtual objects using it. Correct registration improves precision and makes calibration easier [71]. All available sensors, including GPS and compass, determine the positions and surfaces on which items are registered [76]. Registration is likely the most challenging aspect for inexperienced users to evaluate. Therefore, maintaining virtual items in their actual environments after registration is essential, and links should not be easily broken.

(IR) Integration & Real-time data feature: Providing and implementing integration and real-time data, such as predictive preventive maintenance activities, results in realizing the benefits of augmented reality technologies [68]. For instance, AR applications can be integrated with product lifecycle management, IoT, factory execution systems, and enterprise resource planning systems [77] and then provide accurate data from target devices when remote experts are incorporated into augmented reality meetings to guide robust AR system interfaces and sensors [78].

(AD) Architecture & deployment feature: Architectural capability and deployment options are crucial technical characteristics for augmented reality technologies [72]. The system should be user-friendly, dependable, and cost-effective. Thus, the cloud is regarded as an essential component of Industry 4.0 goals. The solution should, therefore, support multiple deployment models, such as cloud, on-premises, and hybrid [79]. Moreover, these systems must be capable of load balancing and high availability [80].

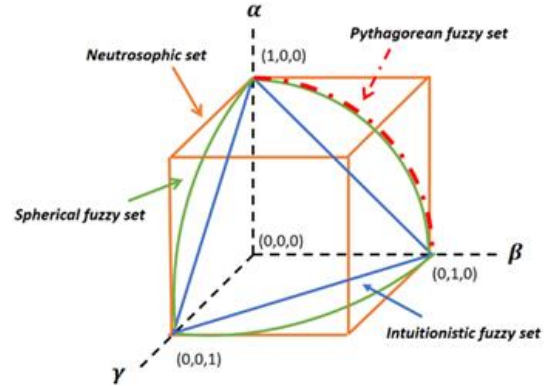
(CA) Cost feature: Cost is unquestionably a crucial aspect of the MCDM procedure, especially when selecting an AR provider [73]. In addition, the cost should include not only the license fee but also maintenance costs, hardware requirements costs, implementation costs, and any expenses incurred due to system-related issues that cannot be resolved by the system [81]. The anticipated return must be sufficient to cover the expense incurred during development and integration, i.e., the investment costs associated with the AR application [82].

III. MATERIALS AND METHODS

A. SPHERICAL FUZZY SETS: BASIC OPERATORS

Mahmood et al. [26] have proposed the most recent extension to the intuitionistic fuzzy set, Pythagorean, neutrosophic, and picture fuzzy sets. Later, Kutlu and Kahraman [83] also introduced SFSs with each Spherical Fuzzy number representing the membership, non-membership, and hesitation functions in the interval [0,1], as presented in **Figure 1**.

FIGURE 1. Geometric representations of SFSs



Definition 1: SFS is denoted as $\tilde{F}_S$ representing a spherical fuzzy set of the universe X as follows:

$$\tilde{F}_S = \{x, (\alpha_{\tilde{F}_S}(x), \beta_{\tilde{F}_S}(x), \gamma_{\tilde{F}_S}(x)) | x \in X\} \quad (1)$$

$$\begin{aligned} \alpha_{\tilde{F}_S}(x): X &\rightarrow [0,1], \beta_{\tilde{F}_S}(x): X \rightarrow [0,1], \\ \gamma_{\tilde{F}_S}(x): X &\rightarrow [0,1] \end{aligned} \quad (2)$$

$$0 \leq \alpha_{\tilde{F}_S}^2(x) + \beta_{\tilde{F}_S}^2(x) + \gamma_{\tilde{F}_S}^2(x) \leq 1 \quad (3)$$

with $\forall x \in X$, $\alpha_{\tilde{F}_S}(x)$ is known as the membership function, $\beta_{\tilde{F}_S}(x)$ is denoted as a non-membership function and $\gamma_{\tilde{F}_S}(x)$ is a hesitant function of x to $\tilde{A}_S$.

Definition 2: Some basic operations of SFS are presented with $\tilde{F}_S = (\alpha_{\tilde{F}_S}, \beta_{\tilde{F}_S}, \gamma_{\tilde{F}_S})$ and $\tilde{E}_S = (\alpha_{\tilde{E}_S}, \beta_{\tilde{E}_S}, \gamma_{\tilde{E}_S})$ as follows:

Union:

$$\begin{aligned} \tilde{F}_S \cup \tilde{E}_S &= \{\max\{\alpha_{\tilde{F}_S}, \alpha_{\tilde{E}_S}\}, \min\{\beta_{\tilde{F}_S}, \beta_{\tilde{E}_S}\}, \\ &\min\left[\sqrt{1 - \left\{(\max\{\alpha_{\tilde{F}_S}, \alpha_{\tilde{E}_S}\})^2 + (\min\{\beta_{\tilde{F}_S}, \beta_{\tilde{E}_S}\})^2\right\}}, \right. \\ &\left. \max\{\gamma_{\tilde{F}_S}, \gamma_{\tilde{E}_S}\}\right] \end{aligned} \quad (4)$$

Intersection:

$$\begin{aligned} \tilde{F}_S \cap \tilde{E}_S &= \{\min\{\alpha_{\tilde{F}_S}, \alpha_{\tilde{E}_S}\}, \max\{\beta_{\tilde{F}_S}, \beta_{\tilde{E}_S}\}, \\ &\max\left[\sqrt{1 - \left\{(\min\{\alpha_{\tilde{F}_S}, \alpha_{\tilde{E}_S}\})^2 + (\max\{\beta_{\tilde{F}_S}, \beta_{\tilde{E}_S}\})^2\right\}}, \right. \\ &\left. \min\{\gamma_{\tilde{F}_S}, \gamma_{\tilde{E}_S}\}\right] \end{aligned} \quad (5)$$

Addition:

$$\tilde{F}_S \oplus \tilde{E}_S = \quad (6)$$

$$\left\{ \begin{array}{l} \sqrt{(\alpha_{\tilde{F}_S}^2 + \alpha_{\tilde{E}_S}^2 - \alpha_{\tilde{F}_S}^2 \alpha_{\tilde{E}_S}^2), \beta_{\tilde{F}_S} \beta_{\tilde{E}_S}}, \\ \sqrt{(1 - \alpha_{\tilde{E}_S}^2) \gamma_{\tilde{F}_S}^2 + (1 - \alpha_{\tilde{F}_S}^2) \gamma_{\tilde{E}_S}^2 - \gamma_{\tilde{F}_S}^2 \gamma_{\tilde{E}_S}^2} \end{array} \right\}$$

Multiplication:

$$\tilde{F}_S \otimes \tilde{E}_S =$$

$$\left\{ \begin{array}{l} \alpha_{\tilde{F}_S} \alpha_{\tilde{E}_S}, \sqrt{(\beta_{\tilde{F}_S}^2 + \beta_{\tilde{E}_S}^2 - \beta_{\tilde{F}_S}^2 \beta_{\tilde{E}_S}^2)}, \\ \sqrt{(1 - \beta_{\tilde{E}_S}^2) \gamma_{\tilde{F}_S}^2 + (1 - \beta_{\tilde{F}_S}^2) \gamma_{\tilde{E}_S}^2 - \gamma_{\tilde{F}_S}^2 \gamma_{\tilde{E}_S}^2} \end{array} \right\}$$

Multiplication by a scalar; $\sigma > 0$:

$$\sigma \cdot \tilde{F}_S = \left\{ \begin{array}{l} \sqrt{1 - (1 - \alpha_{\tilde{F}_S}^2)^\sigma}, \beta_{\tilde{F}_S}^\sigma, \\ \sqrt{(1 - \alpha_{\tilde{F}_S}^2)^\sigma - (1 - \alpha_{\tilde{F}_S}^2 - \gamma_{\tilde{F}_S}^2)^\sigma} \end{array} \right\}$$

Power of $\tilde{F}_S$; $\sigma > 0$

$$\tilde{F}_S^\sigma = \left\{ \begin{array}{l} \alpha_{\tilde{F}_S}^\sigma, \sqrt{(1 - (1 - \beta_{\tilde{F}_S}^2)^\sigma)}, \\ \sqrt{(1 - \beta_{\tilde{F}_S}^2)^\sigma - (1 - \beta_{\tilde{F}_S}^2 - \gamma_{\tilde{F}_S}^2)^\sigma} \end{array} \right\}$$

Definition 3: For these SFSs $\tilde{F}_S = (\alpha_{\tilde{F}_S}, \beta_{\tilde{F}_S}, \gamma_{\tilde{F}_S})$ and $\tilde{E}_S = (\alpha_{\tilde{E}_S}, \beta_{\tilde{E}_S}, \gamma_{\tilde{E}_S})$, the following operations are accepted under the condition $\sigma, \sigma_1, \sigma_2 > 0$:

$$\tilde{F}_S \oplus \tilde{E}_S = \tilde{E}_S \oplus \tilde{F}_S \quad (10)$$

$$\tilde{F}_S \otimes \tilde{E}_S = \tilde{E}_S \otimes \tilde{F}_S \quad (11)$$

$$\sigma(\tilde{F}_S \oplus \tilde{E}_S) = \sigma \tilde{F}_S \oplus \sigma \tilde{E}_S \quad (12)$$

$$\sigma_1 \tilde{F}_S \oplus \sigma_2 \tilde{F}_S = (\sigma_1 + \sigma_2) \tilde{F}_S \quad (13)$$

$$(\tilde{F}_S \otimes \tilde{E}_S)^\sigma = \tilde{F}_S^\sigma \otimes \tilde{E}_S^\sigma \quad (14)$$

$$\tilde{F}_S^{\sigma_1} \otimes \tilde{F}_S^{\sigma_2} = \tilde{F}_S^{\sigma_1 + \sigma_2} \quad (15)$$

Definition 4: Spherical weighted arithmetic mean (SWAM) is calculated using Equation (16):

$$SWAM_w(\tilde{F}_{S1}, \dots, \tilde{F}_{Sn}) = w_1 \tilde{F}_{S1} + w_2 \tilde{F}_{S2} + \dots + w_n \tilde{F}_{Sn} = \quad (16)$$

$$\left\{ \begin{array}{l} \sqrt{\left[1 - \prod_{i=1}^n (1 - \alpha_{\tilde{F}_{Si}}^2)^{w_i}\right]}, \prod_{i=1}^n \beta_{\tilde{F}_{Si}}^{w_i}, \\ \sqrt{\prod_{i=1}^n (1 - \alpha_{\tilde{F}_{Si}}^2)^{w_i} - \prod_{i=1}^n (1 - \alpha_{\tilde{F}_{Si}}^2 - \gamma_{\tilde{F}_{Si}}^2)^{w_i}} \end{array} \right\}$$

$$(7) \quad \text{Where } w = (w_1, w_2, \dots, w_n); \quad w_i \in [0,1]; \quad \sum_{i=1}^n w_i = 1.$$

Definition 5: Spherical weighted geometric mean (SWGM) is calculated using Equation (17):

$$SWGM_w(\tilde{F}_{S1}, \dots, \tilde{F}_{Sn}) = \tilde{F}_{S1}^{w_1} + \tilde{F}_{S2}^{w_2} + \dots + \tilde{F}_{Sn}^{w_n} = \left\{ \begin{array}{l} \prod_{i=1}^n \alpha_{\tilde{F}_{Si}}^{w_i}, \sqrt{1 - \prod_{i=1}^n (1 - \beta_{\tilde{F}_{Si}}^2)^{w_i}}, \\ \sqrt{\prod_{i=1}^n (1 - \beta_{\tilde{F}_{Si}}^2)^{w_i} - \prod_{i=1}^n (1 - \beta_{\tilde{F}_{Si}}^2 - \gamma_{\tilde{F}_{Si}}^2)^{w_i}} \end{array} \right\} \quad (17)$$

$$(9) \quad \text{Where } w = (w_1, w_2, \dots, w_n); \quad w_i \in [0,1]; \quad \sum_{i=1}^n w_i = 1$$

B. PROPOSED METHOD

The proposed method consists of two phases, as shown in **Figure 2**.

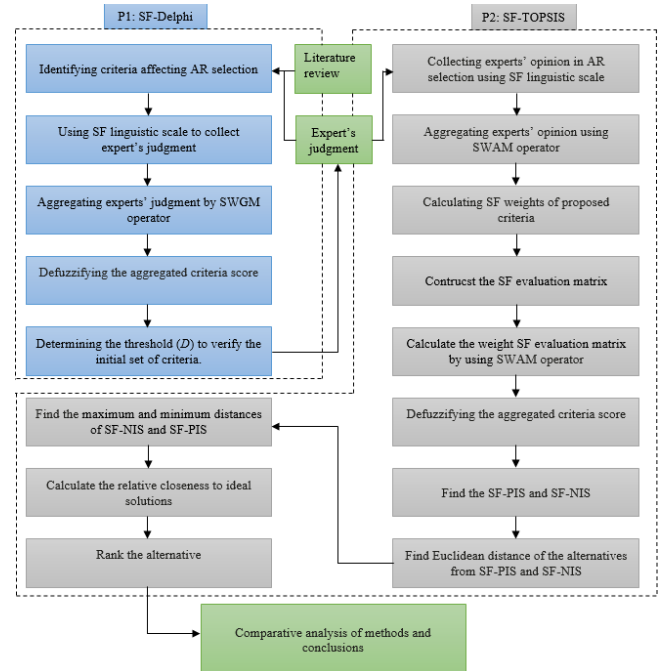


FIGURE 2. The proposed method

Phase I: The SF-Delphi approach is employed to filter the factors. The newly modified SF-Delphi method proposed by Nguyen [60] is represented as follows:

Step 1: decision makers (DMs) Experts must rank the criteria using the linguistic terms listed in **Table 3**.

The SWGM operator is used to obtain the significance vector for each indicator using Equation (17), which is displayed in the Equation as follows:

$$\tilde{J}^{agg} = \begin{bmatrix} \alpha_{11}, \beta_{11}, \gamma_{11} & \cdots & \alpha_{1m}, \beta_{1m}, \gamma_{1m} \\ \vdots & \ddots & \vdots \\ \alpha_{n1}, \beta_{n1}, \gamma_{n1} & \cdots & \alpha_{nm}, \beta_{nm}, \gamma_{nm} \end{bmatrix} \quad (18)$$

Step 2: The Equation calculates the score function:

$$\text{Score}(d_i) = (2\alpha_{ij} - \gamma_{ij})^2 - (\beta_{ij} - \gamma_{ij})^2 \quad (19)$$

Step 3: Validate the list of critical criteria

The threshold is attained as follows:

$$D_i = \sum_{i=1}^n \frac{d_i}{m} \quad (20)$$

If $d_i < D$, criterion C_i is removed, and if $d_i > D$, criterion C_i is valid.

Phase II: Evaluating the alternatives' rankings using the SF-TOPSIS method [60].

TABLE 3. SPHERICAL LINGUISTIC EXPRESSIONS

Linguistic scale	Code	(α, β, γ)
Absolutely more importance	AMI	(0.9,0.1,0.1)
Very high importance	VHI	(0.8,0.2,0.2)
High importance	HI	(0.7,0.3,0.3)
Slightly more importance	SMI	(0.6,0.4,0.4)
Equally importance	EI	(0.5,0.5,0.5)
Slightly low importance	SLI	(0.4,0.6,0.4)
Low importance	LI	(0.3,0.7,0.3)

Step 1: Let DMs fill in the decision matrices using the linguistic terms in **Table 3**. DMs can also evaluate the criteria for their relative importances (weights).

Step 2: Aggregate the judgments of each DM using Equation (16). The aggregative spherical fuzzy decision matrix is constructed using Equation (18).

$$\tilde{D} = (C_j(\tilde{X}_i))_{m \times n} = \begin{pmatrix} (\alpha_{11}, \beta_{11}, \gamma_{11}) & \cdots & (\alpha_{1n}, \beta_{1n}, \gamma_{1n}) \\ (\alpha_{21}, \beta_{21}, \gamma_{21}) & \cdots & (\alpha_{2n}, \beta_{2n}, \gamma_{2n}) \\ \vdots & \ddots & \vdots \\ (\alpha_{m1}, \beta_{m1}, \gamma_{m1}) & \cdots & (\alpha_{mn}, \beta_{mn}, \gamma_{mn}) \end{pmatrix} \quad (21)$$

Step 3: Aggregating the Spherical Fuzzy linguistic evaluations of the decision criteria assigned by decision-makers. Normalize the aggregated criteria weights by using Equation (22). The second way of the complete fuzzy approach is to continue without defuzzifying the criteria weights. Defuzzify the aggregated weighted decision matrix by using the proposed score function using Equation (23).

$$\bar{w}^s_j = \frac{w^s_j}{\sum_{j=1}^n w^s_j} \quad (22)$$

$$\text{Score}(C_j(\tilde{X}_i)) = (\alpha_{ij} - \gamma_{ij})^2 - (\beta_{ij} - \gamma_{ij})^2 \quad (23)$$

Step 4: Determine the **Spherical Fuzzy Positive Ideal Solution (SF-PIS)** and the **Spherical Fuzzy Negative Ideal Solution (SF-NIS)** using Equations (24)-(27).

Equations (24)-(25) are used for the SF-PIS:

$$X^* = \{C_j, \max_i \text{Score}(C_j(\tilde{X}_i)) > | j = 1, 2, \dots, n\} \quad (24)$$

$$\tilde{X}^* = \{ \langle C_1, (\alpha_1^*, \beta_1^*, \gamma_1^*) \rangle, \langle C_2, (\alpha_2^*, \beta_2^*, \gamma_2^*) \rangle \dots \langle C_n, (\alpha_n^*, \beta_n^*, \gamma_n^*) \rangle \} \quad (25)$$

Equations (26)-(27) are used for the SF-NIS:

$$X^- = \{C_j, \min_i \text{Score}(C_j(\tilde{X}_i)) > | j = 1, 2, \dots, n\} \quad (26)$$

$$\tilde{X}^- = \{ \langle C_1, (\alpha_1^-, \beta_1^-, \gamma_1^-) \rangle, \langle C_2, (\alpha_2^-, \beta_2^-, \gamma_2^-) \rangle \dots \langle C_n, (\alpha_n^-, \beta_n^-, \gamma_n^-) \rangle \} \quad (27)$$

Step 5: Measuring the distances of SF-NIS and SF-PIS between alternative X_i using Equation (28)- (29):

$$D(X_i, X^-) = \sqrt{\frac{1}{2n} \sum_{i=1}^n \left((\alpha_{x_i} - \alpha_{x^-})^2 + (\beta_{x_i} - \beta_{x^-})^2 + (\gamma_{x_i} - \gamma_{x^-})^2 \right)} \quad (28)$$

$$D(X_i, X^*) = \sqrt{\frac{1}{2n} \sum_{i=1}^n \left((\alpha_{x_i} - \alpha_{x^*})^2 + (\beta_{x_i} - \beta_{x^*})^2 + (\gamma_{x_i} - \gamma_{x^*})^2 \right)} \quad (29)$$

Step 6: Equations (30)- (31) are applied to obtain maximum and minimum distances of SF-NIS and SF-PIS:

$$D_{\max}(X_i, X^-) = \max_{1 \leq i \leq m} D(X_i, X^-) \quad (30)$$

$$D_{\min}(X_i, X^*) = \min_{1 \leq i \leq m} D(X_i, X^*) \quad (31)$$

Step 7: To obtain zero or positive outcomes, Equation (32) is conducted to calculate the alternatives' ideal ranking order and select the optimal alternative to the increasing values of the closeness ratio:

$$\text{Closeness Ratio}_{X_i}: \quad (32)$$

$$\xi(X_i) = \frac{D(X_i, X^*)}{D_{\min}(X_i, X^*)} - \frac{D(X_i, X^-)}{D_{\max}(X_i, X^-)}$$

IV. CASE STUDY

A. DESCRIPTION

In the initial phase of this academic study, a meticulously chosen panel of experts was assembled, encompassing individuals with diverse backgrounds and expertise directly relevant to Augmented Reality (AR), Information Technology (IT), or the educational domain. The selection criteria focused on robust professional experience, academic credentials, and a demonstrated mastery of the subject matter. While there is no universally stipulated guideline for panel size, the prevailing consensus suggests that a more extensive panel tends to yield more dependable group judgments. Established recommendations propose that each area of expertise ideally consists of a minimum of 10 to 18 panel members [60]. Given the intricacies inherent in the criteria under scrutiny in this study, the author aimed for a minimum sample size of 15 participants.

The survey instrument, meticulously crafted using Google Forms, was emailed to a targeted pool of 20 specialists. From this outreach, a total of 15 responses were received and deemed fit for inclusion in the research. The selected experts each boast a minimum of 10 years of experience in information technology and education and a proven track record of employment in esteemed universities. A comprehensive breakdown of the respondents reveals a balanced gender distribution (10 males, five females), diverse educational backgrounds (7 with a Master's degree, 8 with a Ph.D.), and an average of 12 years of professional experience. The positions held by these experts include 6 University Scholars and 9 Managers, further underscoring the varied and seasoned nature of the participant group. For a more detailed presentation of their profiles, refer to **Table 4**.

TABLE 4. EXPERTS' PROFILE

Gender	Male	10	persons
	Female	5	persons
Education	Master	7	persons

	PhD	8	persons
Experience	Average	12	years
Position	University	6	persons
	Scholars		
	Managers	9	persons

The data collected for this study is analyzed using Microsoft Office Excel. Based on experts' opinions, a set of critical criteria and ten alternatives are considered in this study. Top 10 global AR provider alternatives are reflected as follows: ATLATL Software, Threekit, ZapWorks, CDS Visual, ARitize Events, Dynamics 365 Product Visualize, Epigraph, JigSpace, CanvasLogic, Zakeke, and nine criteria as described in relevant criteria section includes (HS), (CS), (AU), (RR), (TR), (RE), (IR), (AD), and (CA).

B. RESULTS OF SF-DELPHI METHOD

Utilizing spherical linguistics (refer to Table 3), experts were invited to evaluate a list of criteria derived from the literature review, as illustrated in **Table 5**.

TABLE 5. EXPERTS' LINGUISTIC ASSESSMENT

Criteria	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10	E11	E12	E13	E14	E15
HS	HI	AMI	HI	HI	EI	SMI	HI	EI	AMI	HI	HI	HI	HI	HI	HI
CS	HI	VHI	HI	SMI	EI	VHI	HI	HI	AMIVHI	VHI	EI	SMI	VHI	EI	HI
AU	VHI	HI	SMI	SMI	HI	HI	EI	SLI	EI	HI	HI	HI	HI	HI	SLI
RR	HI	EI	VHI	AMISMI	EI	SMI	SMI	HI	VHI	HI	HI	SMI	AMIA	MI	MI
TR	SMI	VHI	HI	SMI	HI	EI	AMI	HI	EI	EI	VHI	HI	VHI	AMIA	MI
RE	AMI	EI	VHI	HI	HI	HI	HI	EI	HI	VHISMI	HI	EI	VHI	VHI	HI
IR	HI	EI	HI	HI	VHI	AMI	HI	VHISMI	AMI	HI	VHI	EI	SMI	EI	HI
AD	AMIVHI	SMI	EI	AMI	HI	VHI	EI	HI	SMI	HI	VHISMI	HI	HI	HI	HI
CA	AMIVHI	EI	HI	EI	AMI	EI	HI	VHI	HI	VHIVHI	HI	HI	EI	HI	HI

This method effectively transformed expert opinions into spherical scales, enabling a more nuanced and precise assessment of the criteria list. Subsequently, the SWGM Operator (Equation (17)) was employed to aggregate the spherical fuzzy numbers representing the opinions of the 15 experts. A threshold value of 0.389 was calculated using Equation (18) to streamline the criteria and focus on the most significant elements. As a result, the criterion AU was removed (refer to **Table 6**). This underscores the importance of meticulously examining and deliberating criteria to identify the most relevant and substantial elements for assessment. Consequently, the proposed method provides a valuable framework for evaluating and selecting AR suppliers based on validated factors.

TABLE 6. THE RESULTS OF THE SF-DELPHI METHOD

Criteria	SFNs(α, β, γ)	Score	Decision
HS	(0.532, 0.406, 0.413)	0.424	Accepted
CS	(0.528, 0.422, 0.428)	0.394	Accepted
AU	(0.444, 0.494, 0.439)	0.198	Rejected
RR	(0.536, 0.411, 0.421)	0.422	Accepted
TR	(0.542, 0.415, 0.424)	0.436	Accepted
RE	(0.529, 0.404, 0.425)	0.401	Accepted
IR	(0.527, 0.418, 0.430)	0.390	Accepted
AD	(0.538, 0.406, 0.416)	0.435	Accepted
CA	(0.533, 0.422, 0.432)	0.402	Accepted

Threshold (D_i)	0.389
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C. RESULTS OF CRITERIA WEIGHTS

The SF-Delphi method yielded eight critical criteria with corresponding weights determined through linguistic evaluation by 15 specialists, detailed in **Table 7**. The choice of the SWAM operator for criteria weight amalgamation is justified by its additivity, linearity, and reduced sensitivity to extreme values, enhancing transparency and robustness. Aligned with decision-making goals emphasizing equitable treatment of all factors, SWAM is well-suited for practical applications, demonstrating simplicity and ease of implementation. Table 8 presents the aggregated spherical fuzzy weight for each criterion using linguistic weights processed through the SWAM operator.

TABLE 7. SPHERICAL LINGUISTIC ASSESSMENT OF CRITERIA

	HS	CS	RR	TR	RE	IR	AD	CA
DM1	LI	EI	AMI	SMI	HI	EI	AMI	ALI
DM2	SLI	VHI	VHI	SMI	SMI	AMI	HI	EI
DM3	HI	AMI	HI	AMI	VHI	SLI	VHI	VLI
DM4	HI	AMI	HI	LI	SMI	AMI	HI	EI
DM5	HI	EI	SMI	EI	VHI	SLI	AMI	AMI
DM6	AMI	HI	VHI	LI	HI	EI	AMI	ALI
DM7	AMI	SLI	VLI	HI	SMI	AMI	HI	EI
DM8	VHI	SLI	SLI	VLI	VHI	SLI	VHI	VLI
DM9	HI	AMI	SMI	SLI	AMI	ALI	HI	EI
DM1 0	VHI	HI	HI	VHI	HI	EI	SMI	AMI
DM1 1	SMI	EI	SMI	LI	VHI	VLI	VHI	SLI
DM1 2	AMI	SLI	LI	HI	HI	EI	SMI	AMI
DM1 3	HI	VHI	VLI	VHI	AMI	AMI	VHI	SLI
DM1 4	EI	VHI	SMI	VHI	AMI	ALI	HI	EI
DM1 5	SMI	EI	HI	SMI	HI	EI	SMI	AMI

TABLE 8. AGGREGATED WEIGHTS OF ALL CRITERIA

Criteria	Spherical fuzzy weights		
HS	0.75	0.26	0.27
CS	0.73	0.28	0.29
RR	0.66	0.36	0.30
TR	0.65	0.37	0.29
RE	0.77	0.23	0.25
IR	0.67	0.37	0.31
AD	0.77	0.23	0.25
CA	0.63	0.41	0.32

D. RESULTS OF SF-TOPSIS METHOD

In further elucidation of the SF-TOPSIS approach, the systematic appraisal and ranking process involves several explicit steps. After determining the criterion weights, the second step of applying the SF-TOPSIS model is to rank the global providers of AR software for education. Initially, the weights of the spherical fuzzy criteria from **Table 8** are multiplied by the criteria of the spherical fuzzy evaluation matrix in **Table 9**, resulting in the weighted spherical fuzzy evaluation matrix presented in **Table 10**.

Subsequently, the score function values are computed, as demonstrated in **Table 11**, using Equation (16). Following this, the weighted spherical fuzzy evaluation matrix undergoes defuzzification to ascertain the spherical fuzzy positive ideal solution (SF-PIS) and spherical fuzzy negative ideal solution (SF-NIS), as detailed in **Table 12**. The Euclidean distances of the alternatives from the SF-PIS (X^*) and SF-NIS (X^-) are then calculated employing Equations (26)–(27), and their relative proximity to the ideal solutions is determined using Equation (32). The results are tabulated in **Table 13** and graphically represented in **Figure 3**, showcasing the distances of the alternatives from the SF-PIS and SF-NIS, their Closeness Ratio (CR) to the Ideal Solutions, and their respective ranks.

TABLE 9. DECISION MATRIX USING SWAM OPERATOR

	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10
HS	0.702	0.560	0.876	0.214	0.707	0.597	0.878	0.237	0.237	0.703
	0.298	0.451	0.127	0.834	0.294	0.418	0.126	0.814	0.814	0.298
	0.300	0.456	0.119	0.188	0.296	0.432	0.116	0.212	0.212	0.300
CS	0.637	0.866	0.722	0.477	0.647	0.865	0.722	0.482	0.482	0.649
	0.366	0.140	0.279	0.531	0.356	0.142	0.279	0.524	0.524	0.354
	0.371	0.136	0.281	0.471	0.364	0.130	0.281	0.472	0.472	0.361

RR	0.809	0.496	0.813	0.515	0.810	0.545	0.812	0.473	0.473	0.810
	0.191	0.524	0.190	0.559	0.190	0.481	0.191	0.596	0.596	0.190
	0.193	0.369	0.190	0.217	0.191	0.351	0.191	0.237	0.237	0.192
TR	0.658	0.866	0.757	0.626	0.654	0.864	0.758	0.599	0.599	0.662
	0.348	0.140	0.246	0.390	0.352	0.143	0.245	0.415	0.415	0.344
	0.361	0.135	0.257	0.422	0.365	0.128	0.257	0.440	0.440	0.357
RE	0.809	0.591	0.900	0.877	0.810	0.603	0.900	0.875	0.875	0.810
	0.191	0.441	0.100	0.125	0.190	0.429	0.100	0.128	0.128	0.190
	0.193	0.331	0.100	0.140	0.191	0.325	0.100	0.137	0.137	0.192
IR	0.707	0.597	0.878	0.237	0.703	0.596	0.878	0.230	0.230	0.698
	0.294	0.418	0.126	0.814	0.298	0.418	0.125	0.821	0.821	0.302
	0.296	0.432	0.116	0.212	0.300	0.437	0.116	0.206	0.206	0.304
AD	0.647	0.865	0.722	0.482	0.649	0.864	0.723	0.481	0.481	0.640
	0.356	0.142	0.279	0.524	0.354	0.143	0.278	0.526	0.526	0.363
	0.364	0.130	0.281	0.472	0.361	0.132	0.280	0.472	0.472	0.369
CA	0.810	0.545	0.812	0.473	0.810	0.541	0.813	0.480	0.480	0.805
	0.190	0.481	0.191	0.596	0.190	0.485	0.190	0.590	0.590	0.195
	0.191	0.351	0.191	0.237	0.192	0.352	0.190	0.234	0.234	0.196

TABLE 10. WEIGHTED DECISION MATRIX BASED ON SWAM OPERATOR

	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10
HS	0.525	0.418	0.655	0.160	0.528	0.446	0.656	0.177	0.177	0.525
	0.390	0.508	0.290	0.846	0.387	0.481	0.289	0.828	0.828	0.390
	0.377	0.485	0.286	0.228	0.374	0.468	0.285	0.250	0.250	0.377
CS	0.465	0.633	0.527	0.348	0.472	0.632	0.527	0.352	0.352	0.474
	0.451	0.314	0.390	0.583	0.444	0.315	0.390	0.577	0.577	0.442
	0.434	0.313	0.379	0.496	0.429	0.311	0.379	0.497	0.497	0.428
RR	0.530	0.325	0.533	0.338	0.531	0.357	0.532	0.310	0.310	0.531
	0.405	0.609	0.404	0.635	0.405	0.577	0.405	0.664	0.664	0.405
	0.340	0.414	0.339	0.314	0.340	0.406	0.339	0.319	0.319	0.340
TR	0.429	0.564	0.494	0.408	0.426	0.563	0.494	0.391	0.391	0.432
	0.492	0.394	0.436	0.519	0.495	0.395	0.436	0.535	0.535	0.490
	0.420	0.314	0.364	0.460	0.423	0.312	0.364	0.471	0.471	0.419
RE	0.626	0.457	0.697	0.679	0.627	0.467	0.697	0.677	0.677	0.627
	0.296	0.487	0.250	0.260	0.295	0.477	0.250	0.262	0.262	0.295
	0.303	0.383	0.264	0.279	0.303	0.379	0.264	0.277	0.277	0.303
IR	0.472	0.399	0.587	0.158	0.469	0.398	0.587	0.154	0.154	0.467
	0.461	0.537	0.389	0.842	0.463	0.537	0.389	0.848	0.848	0.465
	0.394	0.472	0.325	0.259	0.397	0.476	0.325	0.253	0.253	0.398
AD	0.498	0.666	0.556	0.371	0.500	0.665	0.556	0.370	0.370	0.492
	0.418	0.272	0.358	0.561	0.416	0.272	0.358	0.562	0.562	0.423
	0.414	0.276	0.357	0.492	0.412	0.277	0.357	0.492	0.492	0.418
CA	0.514	0.345	0.514	0.300	0.513	0.343	0.515	0.304	0.304	0.510
	0.446	0.601	0.447	0.681	0.446	0.604	0.446	0.677	0.677	0.448
	0.356	0.411	0.355	0.328	0.356	0.412	0.355	0.328	0.328	0.357

TABLE 11. SCORE FUNCTION VALUES

Alternatives	HS	CS	RR	TR	RE	IR	AD	CA
X1	0.022	0.001	0.032	-0.005	0.104	0.002	0.007	0.017
X2	0.004	0.102	-0.030	0.056	-0.005	0.001	0.152	-0.032
X3	0.136	0.022	0.033	0.012	0.187	0.064	0.040	0.017
X4	-0.378	0.014	-0.103	-0.001	0.160	-0.330	0.010	-0.124
X5	0.024	0.002	0.033	-0.005	0.105	0.001	0.008	0.017
X6	0.000	0.103	-0.027	0.056	-0.002	0.002	0.151	-0.032
X7	0.138	0.022	0.033	0.012	0.187	0.064	0.040	0.017
X8	-0.329	0.015	-0.119	0.002	0.160	-0.343	0.010	-0.122
X9	-0.329	0.015	-0.119	0.002	0.160	-0.343	0.010	-0.122
X10	0.022	0.002	0.032	-0.005	0.105	0.000	0.006	0.015

TABLE 12. SF PIS AND SF NIS

Criteria	X*	X-	Criteria	X*	X-
HS	0.656	0.160	RE	0.697	0.457
	0.289	0.846		0.250	0.487
	0.285	0.228		0.264	0.383
CS	0.632	0.465	IR	0.587	0.154
	0.315	0.451		0.389	0.848
	0.311	0.434		0.325	0.253
RR	0.533	0.310	AD	0.666	0.492
	0.404	0.664		0.272	0.423
	0.339	0.319		0.276	0.418
TR	0.563	0.426	CA	0.515	0.300
	0.395	0.495		0.446	0.681
	0.312	0.423		0.355	0.328

TABLE 13. RESULTS OF ALTERNATIVE RANKING

Alternatives	PIS D (X _i , X*)	NIS D (X _i , X-)	Closeness Ratio (CR)	Rank
X1	0.122	0.240	1.226	5
X2	0.176	0.202	2.282	7
X3	0.059	0.290	0.004	2
X4	0.317	0.109	4.989	8
X5	0.120	0.241	1.197	3
X6	0.165	0.207	2.072	6
X7	0.059	0.290	0	1
X8	0.318	0.109	4.994	9
X9	0.318	0.109	4.994	9
X10	0.121	0.239	1.221	4

The closeness ratios imply that Epigraph is the most suitable global AR supplier for education. The priority ranking of the global AR supplier within the scope of our research is $X7 > X3 > X4 > X10 > X1 > X6 > X2 > X4 > X8 > X9$. Epigraph, ATLATL Software, and CDS Visual are the top three potential global AR suppliers, with ratings of 0, 0.04, and 1.197, respectively.

It is well-noted that The novel spherical fuzzy TOPSIS model provides a powerful tool for managers and decision-makers to make strategic decisions and produce reliable

results. Managers in various industries can use this framework for various applications, including supplier selection, strategy selection, etc. However, this framework's decision-making processes will depend on the preferences of experts and the individuals involved in the decision-making process. A comparison is also conducted to determine the result's relevance and reliability. Thus, managers and decision-makers must be aware of the model's limitations and the potential biases that may arise from the preferences of experts and individuals.

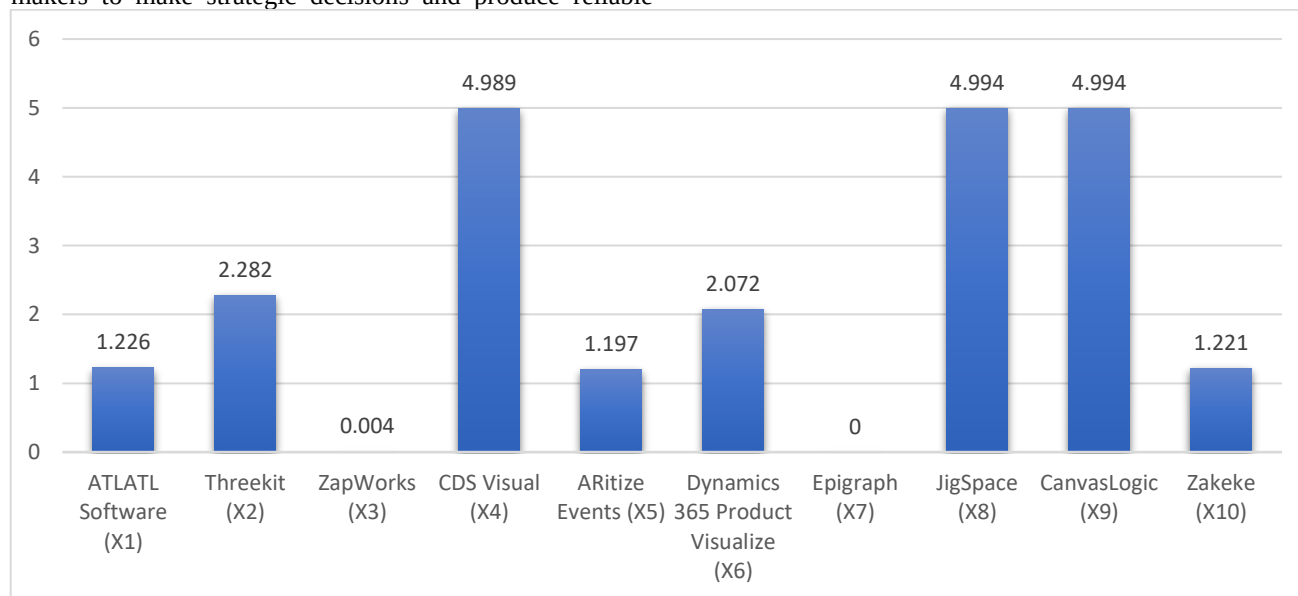


FIGURE 3. Final ranking of alternative

E. COMPARATIVE ANALYSIS

To ensure that the MCDM approach is practical, it is necessary to demonstrate the applicability and rationality of the proposed methods by comparing them to established methods commonly used in related studies. This comparison should be conducted methodically, considering both the advantages and disadvantages of the proposed method and the established methods and the practicality and potential for success of the proposed method in the given context [83]. In this study, the ranking of global AR providers for education using the SF-TOPSIS model is evaluated by comparing it with two other established methods: WASPAS and MULTIMOORA. It is noted that the WASPAS method, which was first introduced in 2012, is well-known and practical for resolving multi-criteria decision-making problems [84]. It combines two approaches: the weighted sum model (WSM) and the weighted product model (WPM), which provides both benefits. The WSM method is simple,

while the WPM approach helps avoid solutions with low values.

Moreover, the multi-objective optimization based on ratio analysis (MOORA) method, proposed by Brauers and Zavadskas [85], has uncomplicated mathematics, making it effortless and practical for selecting materials. The MULTIMOORA method, which is a comprehensive form of the MOORA technique, was introduced by Brauers and Ginevičius in 2010. The final rank generated by the MULTIMOORA technique is more robust than traditional MADM methods since it integrates three subordinate ranks [86].

Table 14 and Figure 4 present the comparison of the three ranking methods. The results indicate that the ranking of global AR suppliers for education obtained from the proposed SF-TOPSIS model is similar to those obtained

from the SF-WASPAS and SF-MULTIMOORA models. However, there are minor variations in the rankings obtained from these methods. In the SF-WASPAS model, the difference is observed between ATLATL Software (X1) and Zakeke (X10). On the other hand, only ATLATL Software

(X2), ARitize Events (X5), and Dynamics 365 Product Visualize (X6) had unchanged ratings compared to the proposed model in the MULTIMOORA model. The ratings for the other vendors experienced slight changes.

TABLE 14. THE COMPARISON OF THREE KINDS OF RANKING

Code	Provider	SF-TOPSIS		SF-WASPAS		SF-MULTIMOORA	
		Value	Ranking	Value	Ranking	Value	Ranking
X1	ATLATL Software	1.225706	5	0.217851	4	0.140738	5
X2	Threekit	2.282399	7	0.16534	7	0.29095	7
X3	ZapWorks	0.003957	2	0.434891	2	0	1
X4	CDS Visual	4.988657	8	0.064165	8	0.532853	10
X5	ARitize Events	1.197099	3	0.219431	3	0.136906	3
X6	Dynamics 365 Product Visualize	2.072233	6	0.17473	6	0.257449	6
X7	Epigraph	0	1	0.436056	1	0.00108	2
X8	JigSpace	4.993815	9	0.056506	9	0.509413	8
X9	CanvasLogic	4.993815	9	0.056506	9	0.509413	8
X10	Zakeke	1.221327	4	0.216438	5	0.140648	4

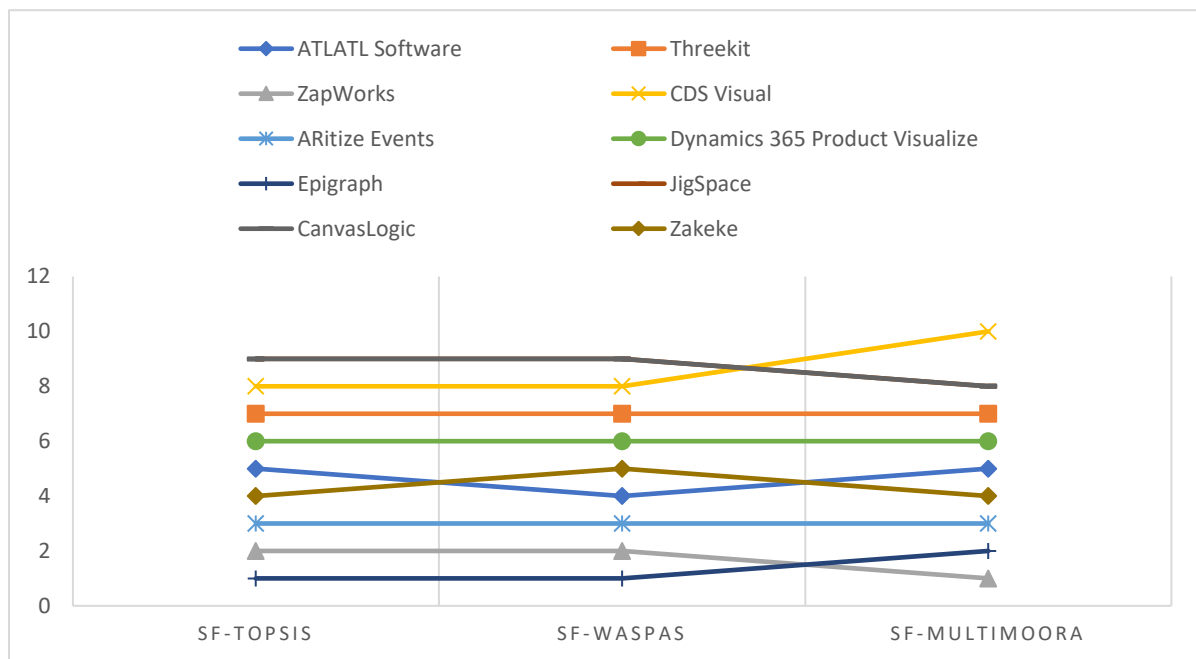


FIGURE 4. Ranking of alternatives compared to different methods

V. CONCLUSION AND IMPLICATIONS

A. CONCLUSION

In conclusion, this study addresses the need for a comprehensive and efficient approach to selecting AR providers in the education sector. With the increasing popularity of AR and the shift towards remote learning due to the COVID-19 pandemic, there is a growing demand for

innovative and effective AR solutions. The proposed decision support model based on SFSs provides a systematic and objective framework for AR provider selection. The model ensures a well-rounded evaluation process by considering the relative importance of decision-makers and aggregating criteria weights through the SWAM operator. The SF-Delphi method identifies and refines eight criteria

that impact AR provider selection. The SF-TOPSIS approach then allows for evaluating ten global AR providers, facilitating a comparative analysis between the proposed SF-TOPSIS model and SF-WASPAS and SF-MULTIMOORA that validates the model's capability and robustness. By streamlining the selection process and providing decision-makers with a time-efficient method, the proposed framework assists in identifying the most suitable AR software solutions. This contributes to enhancing the absorption of learning information by leveraging the creative potential of AR technology in educational settings. Moreover, the findings of this study offer valuable insights into the dynamics of AR applications in the education sector. They shed light on the benefits and potential of AR and encourage innovative utilization of this influential technology to enhance learning experiences.

B. IMPLICATIONS

Theoretical Implications

This study contributes to the theoretical understanding of AR technology adoption in the education sector and the decision-making process for selecting AR providers. Regarding Spherical Fuzzy Sets, this study illustrates the efficacy of SFSs in managing uncertainty and imprecision in decision-making, providing a more robust and flexible approach to AR provider selection. Simultaneously, developing a decision support model customized for AR provider selection enriches the literature on decision-making within technology adoption. The model offers a structured framework that seamlessly integrates multiple criteria and accommodates decision-maker preferences, enhancing the theoretical understanding of decision-making processes in intricate technological environments. Also, the SF-Delphi method aids in defining and refining evaluation criteria through expert consensus, while the SF-TOPSIS approach facilitates comparative analysis and decision-making based on preference similarity to an ideal solution. These methodologies can be adapted and extended to other technology adoption contexts.

Managerial Implications

The managerial implications drawn from this study provide crucial guidance for decision-makers and managers in the education sector contemplating adopting AR solutions. The proposed decision support model not only equips managers with a systematic and objective framework for selecting AR providers but also streamlines the decision-making process, ensuring informed choices aligned with specific institutional requirements and goals. This structured approach significantly expedites the AR provider selection process, resulting in tangible time and resource savings

advantages. By emphasizing customization and branding, managers can leverage insights to select AR providers offering flexibility, enabling the creation of unique and tailored AR experiences that enhance the institution's image. The study underscores the importance of AR providers with robust integration capabilities, facilitating efficient data exchange, personalized learning experiences, and streamlined administrative processes. Moreover, the outlined cost considerations assist managers in making informed and cost-effective decisions, optimizing return on investment. Ultimately, the study positions the adoption of AR technology and effective integration into educational practices as a strategic move, allowing educational institutions to emerge as leaders in technological innovation, attract students, and enhance overall competitiveness in the dynamic educational landscape.

C. LIMITATIONS AND FUTURE RESEARCH

Despite the valuable insights gleaned from this study, it is imperative to acknowledge its limitations. Firstly, the study's findings are rooted in a specific educational context, and caution is warranted when generalizing these results to other educational institutions or sectors, as different settings may present unique requirements and constraints influencing the AR provider selection process differently. Secondly, the aggregation of fixed criteria weights in the decision support model relies on the subjective judgment of decision-makers. Despite employing the SF-Delphi method to ensure consensus, the inherent subjectivity limits the objectivity of the decision-making process. Thirdly, the study identified and included eight criteria for AR provider selection based on expert consensus, yet additional criteria impacting the decision-making process that was not considered may exist.

Additionally, while the study evaluated ten global AR providers, it may not encompass the entire spectrum of available AR solutions. Regional or niche providers with unique offerings were omitted, limiting the representation of the diversity of AR providers worldwide. Finally, it is essential to note that the study did not conduct sensitivity analysis by adjusting criteria weights, which could have provided insights into the robustness of the results under varying weight scenarios. This limitation underscores the importance of future research exploring the impact of changes in criteria weights on the AR provider selection outcomes.

Based on the limitations identified, several avenues for future research can be explored:

Firstly, conducting similar studies in different educational settings and contexts would provide a more comprehensive understanding of the factors influencing AR provider selection. Comparing the results across various contexts

would shed light on the contextual variations and their impact on decision-making processes. Secondly, further research can focus on identifying additional criteria relevant to AR provider selection. This would involve gathering insights from a broader range of stakeholders, including educators, students, and IT professionals, to ensure a comprehensive and diverse set of criteria reflecting AR adoption's multifaceted nature. Thirdly, exploring alternative methods for aggregating criteria weights could enhance the objectivity of the decision support model. Future research could investigate the use of data-driven techniques, or alternative decision-making approaches that minimize the influence of individual biases and provide a more rigorous and transparent weighting process. Fourthly, conducting longitudinal studies that track the performance and outcomes of selected AR providers over an extended period would provide valuable insights into their long-term effectiveness and scalability.

DATA AVAILABILITY

This article includes all the data generated or analyzed during the study.

CONFLICTS OF INTEREST

The author declares that they have no conflicts of interest.

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